

## KERCH, Ukraine

WMO: 339830

Lat: 45.3742N

Lon: 36.4147E

Elev: 161

StdP: 14.61

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.6       | 17.5 | 8.3                           | 8.5 | 13.9 | 12.3 | 10.4 | 18.6 | 31.5                  | 29.2 | 27.9 | 28.6 | 11.7                  | 60   | 0.560 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.5                   | 89.8            | 72.1 | 87.4 | 71.3 | 85.1 | 70.5 | 75.2                | 84.9 | 73.7 | 83.2 | 72.3 | 81.7 | 11.0                 | 30   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.2                            | 119.8 | 80.8 | 70.6 | 113.4 | 79.3 | 69.1 | 107.7 | 78.2 | 38.8          | 84.8 | 37.4 | 83.0 | 36.2 | 81.9 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 26.1          | 22.7 | 19.3 | DB                         | 8.9 | 93.2               | 7.0 | 3.0                                                | 3.9 | 95.4       | -0.2 | 97.1       | -4.1 | 98.8       | -9.2 | 100.9 | (4) |
| (5)               |      |      | WB                         | 8.5 | 78.2               | 6.7 | 2.3                                                | 3.6 | 79.8       | -0.3 | 81.2       | -4.1 | 82.5       | -9.0 | 84.2  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 53.6        | 33.2  | 34.4  | 40.6  | 50.5  | 61.2  | 69.9  | 75.2  | 75.0  | 65.4  | 54.5  | 44.7  | 37.4  | (6)  |  |
|                                                                        | DBStd       | 16.47       | 8.90  | 9.05  | 6.69  | 5.85  | 6.05  | 5.29  | 4.55  | 5.26  | 6.03  | 7.15  | 7.55  | 8.44  | (7)  |  |
|                                                                        | HDD50       | 1942        | 520   | 437   | 296   | 61    | 2     | 0     | 0     | 0     | 0     | 37    | 193   | 396   | (8)  |  |
|                                                                        | HDD65       | 5065        | 985   | 856   | 755   | 435   | 149   | 17    | 0     | 2     | 67    | 333   | 610   | 856   | (9)  |  |
|                                                                        | CDD50       | 3262        | 0     | 1     | 5     | 77    | 349   | 596   | 782   | 775   | 463   | 176   | 33    | 5     | (10) |  |
|                                                                        | CDD65       | 911         | 0     | 0     | 0     | 0     | 31    | 163   | 317   | 312   | 81    | 7     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 7281        | 0     | 0     | 0     | 4     | 152   | 1098  | 2834  | 2773  | 406   | 14    | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 2157        | 0     | 0     | 0     | 0     | 11    | 242   | 903   | 933   | 67    | 1     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 10.1        | 10.8  | 11.4  | 11.4  | 10.2  | 9.0   | 9.0   | 9.4   | 9.6   | 9.7   | 9.9   | 9.7   | 10.7  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 15.9        | 1.1   | 1.2   | 1.1   | 1.5   | 1.1   | 2.0   | 1.7   | 1.8   | 1.5   | 1.0   | 1.5   | 1.6   | (15) |  |
|                                                                        | PrecMax     | 23.7        | 2.7   | 3.2   | 3.0   | 4.3   | 4.1   | 6.1   | 5.6   | 5.5   | 6.1   | 3.2   | 3.3   | 3.9   | (16) |  |
|                                                                        | PrecMin     | 10.0        | 0.2   | 0.5   | 0.0   | 0.2   | 0.0   | 0.2   | 0.0   | 0.0   | 0.1   | 0.1   | 0.0   | 0.1   | (17) |  |
|                                                                        | PrecStd     | 3.3         | 0.7   | 0.7   | 0.7   | 0.9   | 0.8   | 1.5   | 1.7   | 1.6   | 1.5   | 0.7   | 0.9   | 0.9   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 52.9  | 55.0  | 61.0  | 72.4  | 81.4  | 88.8  | 92.4  | 93.6  | 86.2  | 75.6  | 65.7  | 58.4  | (19) |  |
|                                                                        |             | MCWB        | 48.6  | 48.8  | 50.8  | 58.9  | 67.4  | 71.5  | 72.9  | 73.5  | 69.5  | 64.5  | 58.4  | 53.8  | (20) |  |
|                                                                        | 2%          | DB          | 50.1  | 51.2  | 56.5  | 67.5  | 78.0  | 85.2  | 89.8  | 90.3  | 81.5  | 71.9  | 61.7  | 54.7  | (21) |  |
|                                                                        |             | MCWB        | 46.9  | 47.0  | 48.7  | 56.5  | 65.7  | 70.2  | 72.4  | 72.2  | 68.0  | 63.1  | 56.7  | 51.5  | (22) |  |
|                                                                        | 5%          | DB          | 47.8  | 48.5  | 53.1  | 63.6  | 75.1  | 82.8  | 87.4  | 87.8  | 78.4  | 68.4  | 58.4  | 52.1  | (23) |  |
|                                                                        |             | MCWB        | 45.4  | 45.2  | 47.0  | 54.5  | 64.3  | 69.4  | 71.7  | 71.2  | 66.3  | 61.8  | 54.6  | 49.5  | (24) |  |
|                                                                        | 10%         | DB          | 45.6  | 46.1  | 50.2  | 60.3  | 72.3  | 80.2  | 85.2  | 85.3  | 75.7  | 65.3  | 55.6  | 49.7  | (25) |  |
|                                                                        |             | MCWB        | 43.4  | 43.4  | 45.4  | 52.5  | 62.8  | 67.9  | 70.9  | 70.4  | 64.9  | 59.7  | 52.3  | 47.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 49.4  | 50.0  | 52.1  | 61.3  | 70.2  | 75.3  | 77.4  | 77.4  | 73.1  | 67.1  | 59.6  | 55.1  | (27) |  |
|                                                                        |             | MCDB        | 51.9  | 53.5  | 59.7  | 70.2  | 78.9  | 84.4  | 86.9  | 87.6  | 81.9  | 73.0  | 64.0  | 56.9  | (28) |  |
|                                                                        | 2%          | WB          | 47.2  | 47.6  | 49.6  | 57.5  | 67.3  | 73.0  | 75.4  | 75.1  | 70.2  | 64.7  | 57.2  | 51.6  | (29) |  |
|                                                                        |             | MCDB        | 49.6  | 50.6  | 54.8  | 65.4  | 75.5  | 81.9  | 84.9  | 85.1  | 78.3  | 69.3  | 60.9  | 53.9  | (30) |  |
|                                                                        | 5%          | WB          | 45.6  | 45.6  | 47.8  | 55.3  | 65.5  | 71.1  | 73.8  | 73.6  | 68.4  | 62.7  | 55.1  | 49.6  | (31) |  |
|                                                                        |             | MCDB        | 47.8  | 48.2  | 52.1  | 62.5  | 73.1  | 79.7  | 83.4  | 83.3  | 75.7  | 67.3  | 58.1  | 52.0  | (32) |  |
|                                                                        | 10%         | WB          | 43.5  | 43.5  | 46.0  | 53.3  | 63.7  | 69.5  | 72.4  | 72.3  | 66.5  | 60.3  | 52.6  | 47.4  | (33) |  |
|                                                                        |             | MCDB        | 45.5  | 45.9  | 49.7  | 58.7  | 70.9  | 77.6  | 81.9  | 81.7  | 73.4  | 64.7  | 55.4  | 49.7  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.1   | 10.0  | 11.7  | 14.2  | 15.3  | 15.0  | 15.5  | 15.1  | 14.8  | 12.6  | 10.6  | 9.0   | (35) |  |
|                                                                        |             | MCDBR       | 11.4  | 12.5  | 16.4  | 19.8  | 18.9  | 17.9  | 17.7  | 17.8  | 17.9  | 16.7  | 14.4  | 12.3  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 9.3   | 9.2   | 10.4  | 11.2  | 9.8   | 8.2   | 7.7   | 7.4   | 9.0   | 9.9   | 10.1  | 9.8   | (37) |  |
|                                                                        |             | MCWBR       | 11.4  | 11.6  | 15.0  | 17.6  | 17.4  | 15.8  | 16.0  | 16.3  | 16.6  | 15.5  | 13.8  | 11.9  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.352 | 0.382 | 0.411 | 0.447 | 0.414 | 0.418 | 0.440 | 0.443 | 0.414 | 0.387 | 0.360 | 0.351 | (40) |  |
|                                                                        |             | taud        | 2.314 | 2.236 | 2.179 | 2.123 | 2.323 | 2.353 | 2.302 | 2.279 | 2.323 | 2.348 | 2.358 | 2.315 | (41) |  |
|                                                                        | Ebn at Noon |             | 232   | 247   | 257   | 258   | 272   | 270   | 262   | 257   | 254   | 243   | 228   | 218   | (42) |  |
|                                                                        |             | Edh at Noon | 26    | 33    | 40    | 47    | 39    | 38    | 40    | 39    | 35    | 30    | 25    | 23    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 336   | 587   | 1007  | 1549  | 1961  | 2116  | 2122  | 1867  | 1365  | 835   | 449   | 293   | (44) |  |
|                                                                        | RadStd      |             | 45    | 85    | 92    | 143   | 152   | 123   | 94    | 122   | 96    | 85    | 53    | 36    | (45) |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |      |
| (46) | Station Only           | +1.31 | N/A     | N/A    | +2.00   | N/A   | N/A   | -247        | -364  | +320  | +227  | (46) |
| (47) | Regional (5 neighbors) | +1.28 | N/A     | N/A    | N/A     | N/A   | N/A   | -212        | -308  | +281  | +167  | (47) |

Nomenclature: See separate page